



Aerospace Academy Version

Unit 3

CAD Modeling for Aerospace Components

Unit Overview

Students learn to use CAD as an engineering tool by creating parametric sketches, applying constraints, building 3D features, inspecting models, and documenting an aerospace component.

Essential Question

How do engineers turn sketches into precise digital models that can be tested, revised, and manufactured?

Name: _____ Period: _____ Date: _____

Learning Targets

1. Explain the major concepts of cad modeling for aerospace components.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

Unit Deliverables

- Fusion interface notes
- Parametric sketch practice
- Core feature practice
- Aerospace component model
- Model inspection checklist
- Screenshot documentation
- Exported file or drawing

Vocabulary

Term	Definition
CAD	
Parametric Modeling	
Sketch Plane	
Constraint	
Dimension	
Extrude	
Revolve	
Fillet	
Chamfer	
Shell	
Pattern	
Component	
Browser	
Timeline	
Inspection	
Export	

Unit Project: CAD Aerospace Component Project

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

Day 1 — CAD as an Engineering Tool

Objective: I can explain why engineers use CAD.

Bellwork: What can CAD do that a hand sketch cannot?

Core Task: Explore how CAD supports precision, revision, simulation, and manufacturing.

Evidence of Work / Notes:

Exit Ticket: Why is CAD more than drawing on a computer?

Day 2 — Fusion Interface

Objective: I can identify key parts of the Fusion workspace.

Bellwork: What tools or menus do you expect in CAD software?

Core Task: Identify browser, timeline, sketch tools, origin, view cube, and feature tools.

Evidence of Work / Notes:

Exit Ticket: What interface feature will you use most?

Day 3 — Sketch Planes and Profiles

Objective: I can start a CAD sketch on the correct plane.

Bellwork: Why does the starting plane matter?

Core Task: Create simple 2D profiles on different planes.

Evidence of Work / Notes:

Exit Ticket: What plane did you use and why?

Day 4 — Dimensions and Constraints

Objective: I can fully define simple sketches.

Bellwork: Why should sketches be controlled with dimensions?

Core Task: Apply dimensions and geometric constraints to simple profiles.

Evidence of Work / Notes:

Exit Ticket: What constraint did you use today?

Day 5 — Extrude and Cut

Objective: I can use extrude features to add or remove material.

Bellwork: How does a 2D profile become a 3D part?

Core Task: Create a basic bracket using extrude and extrude cut.

Evidence of Work / Notes:

Exit Ticket: What feature removed material?

Day 6 — Fillet, Chamfer, and Shell

Objective: I can use common features to improve part geometry.

Bellwork: Why are sharp edges often modified?

Core Task: Add fillets, chamfers, and shell features to a practice part.

Evidence of Work / Notes:

Exit Ticket: How did one feature change the part?

Day 7 — Patterns and Symmetry

Objective: I can use patterns or mirrored features efficiently.

Bellwork: Why should engineers avoid redrawing repeated features?

Core Task: Create repeated holes, ribs, or slots using pattern tools.

Evidence of Work / Notes:

Exit Ticket: How did a pattern save time?

Day 8 — Aerospace Component Planning

Objective: I can plan a CAD component before modeling.

Bellwork: What should be decided before opening CAD?

Core Task: Select a component: sensor mount, fin bracket, payload case, wing rib, or control horn.

Evidence of Work / Notes:

Exit Ticket: What component will you model?

Day 9 — Component Modeling Day 1

Objective: I can create the base geometry for my component.

Bellwork: What is the first feature of your part?

Core Task: Model the base shape using sketches, dimensions, and extrudes.

Evidence of Work / Notes:

Exit Ticket: What dimension controls your design?

Day 10 — Component Modeling Day 2

Objective: I can add functional details to my CAD model.

Bellwork: What features help your part complete its function?

Core Task: Add holes, slots, ribs, cutouts, mounts, or clearance features.

Evidence of Work / Notes:

Exit Ticket: What functional feature did you add?

Day 11 — Model Inspection

Objective: I can inspect a model for accuracy and design intent.

Bellwork: How can a CAD model be wrong even if it looks good?

Core Task: Check dimensions, constraints, feature order, symmetry, and manufacturability.

Evidence of Work / Notes:

Exit Ticket: What issue did you find?

Day 12 — Revision and Version Control

Objective: I can revise a CAD model and save versions.

Bellwork: Why should engineers keep versions?

Core Task: Revise the model based on inspection and save a new version.

Evidence of Work / Notes:

Exit Ticket: What changed from V1 to V2?

Day 13 — Screenshots and Documentation

Objective: I can document a CAD model clearly.

Bellwork: What screenshots help communicate a part?

Core Task: Capture isometric and orthographic screenshots with notes.

Evidence of Work / Notes:

Exit Ticket: What view best shows your part?

Day 14 — Exporting Files

Objective: I can export files for manufacturing or documentation.

Bellwork: Why do different machines require different file types?

Core Task: Practice exporting STL, drawing PDF, or image files as directed.

Evidence of Work / Notes:

Exit Ticket: What file type did you export?

Day 15 — CAD Component Submission

Objective: I can submit and reflect on a complete CAD component model.

Bellwork: What makes a CAD model professional?

Core Task: Submit model documentation and complete reflection.

Evidence of Work / Notes:

Exit Ticket: What CAD skill improved most?

Unit 3 Submission Checklist

Item	Complete?
Fusion interface notes	☐
Parametric sketch practice	☐
Core feature practice	☐
Aerospace component model	☐
Model inspection checklist	☐
Screenshot documentation	☐
Exported file or drawing	☐

Unit 3 Project Rubric

Category	Points
Technical accuracy and completeness	20
Engineering process and documentation	20
Aerospace connection and design intent	15
Quality of sketches, models, data, or drawings	20
Use of evidence, testing, or decision making	15
Reflection and professionalism	10
Total	100

Unit 3 Student Work Templates

CAD Planning Sketch

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name: _____

Scale: _____

Purpose: _____

Date: _____



Notes / annotations: _____

Parametric Design Plan

Purpose: CAD planning

Directions: List the important dimensions and constraints that should control your model.

Dimension / Parameter	Purpose	Value / Rule
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Modeling sequence:

Model Inspection Checklist

Purpose: Quality control

Directions: Use this before exporting or submitting a CAD model.

Item	Complete?
File named correctly	☐
Sketches are constrained	☐
Important dimensions are intentional	☐
Features are named or organized	☐
Edges are finished appropriately	☐
Model matches project requirements	☐
Screenshot or render exported	☐
Reflection completed	☐

Inspection notes: